10PB24MATS01 MS

Pre-Board Examination- Class 10
MATHEMATICS (Standard)

Marking Scheme

Section A

For MCQ (1 to 20) correct answer l mark each.

1. (a) 2
2. (c) 140°
3. (b) 5 : 6
4. (b) (36^o)
5. (d) four times
6. (c) isosceles right angled triangle
7. (d) -2, -6
8. (d) infinitely many
9. (c) 3 : 1
10. (a) 25
11. (b) 4
12. (b) 8.67
13. (b) 115°
14. (a) -32
15. (c) $4/11$
16. (b) similar but not congruent
17. (b) 2
18. (b) only (i) and (ii)
19. (c) Assertion (A) is True but Reason (R) is false.
20. (b) Both Assertion (A) and Reason (R) are True but Reason (R) is not the correct explanation of Assertion (A)

Section B

21. For finding correct values of x and y using any method, x = 2 and y = 3 : 2 marks

OR

For finding correct values of x and y using any method,

$x = 5.5$ and $y = 4.5$: 2 marks

22. $(r_1/r_2) = ¾$: 1 mark

$(r_1/r_2)^2 = 9/16$

Ratio of the surface area is 9/16 : 1 mark

23. For neat diagram, Given and To prove : 1 mark

For correct proof : 1 mark

24. Area of the rectangle = 20 X 16 $\Longrightarrow$ 320 m^2 : ½ marks

Area that the cow can graze = $(1/4)\pi r^2 \Rightarrow 154\ m^2$

Area that cow cannot graze = 166 m^2 : 1½ marks

25. Sum of zeros = 6

Product of zeros = 6 : 1 mark

$p(x) = x^2 - 6x + 6$: 1 mark

Section C

26. $(\sin A - 2\sin^3 A)/(2\cos^3 A - \cos A)$

$= \sin A(1 - 2\sin^2 A) / \cos(2\cos^2 A - 1)$: 1 mark

$= \sin A(1 - 2\sin^2 A) / \cos(2(1 - \sin^2 A) - 1)$: 1 mark

$= \sin A / \cos A = \tan A$: 1 mark

27. Let us assume that $1 - 2\sqrt{3}$ is rational

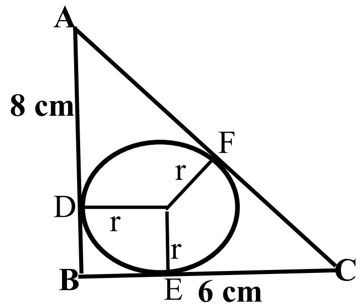
Then we can find a and b such that $1 - 2\sqrt{3} = (a/b)$ where $b \neq 0$ and assume that a and b are co-prime. :1 mark

$1 - (a/b) = 2\sqrt{3}$ or $(b - a) / 2b = \sqrt{3}$

It is not possible to a rational LHS and an irrational RHS. Hence $1 - 2\sqrt{3}$ is irrational : 2 marks

28. (i). $9/10$ (ii). $1/10$ (iii). $1/5$: 1 mark each

29. AC = 10 cm . Let BE = x



AD =AF; BD = BE; CF=CE : 1 mark

AD = 8 – x and EC = 6 – x

Ie. 8 – x + 6 – x = 10

x = 2. Hence the radius = 2 cm. : 2 marks

30. Let x be the fixed charge and y be the daily charge after 3 days.

In the case of Priya, the equation is $x + 5y = 50$.

In the case of Shilpa, the equation is $x + 10y = 85$. : 1½ marks

On solving for x and y, $x = 15$ and $y = 7$.

The fixed charge is ₹15 and daily charge is ₹7. :1½ marks

OR

Let x be the numerator and y be the denominator.

$x = y - 4 \longrightarrow$ (Eq. 1); $8(x - 2) = y + 1 \longrightarrow$ (Eq. 2). : 1½ marks

solving for x and y, $x = 3$; $y = 7$.

The fraction is 3/7 : 1½ marks

31. Volume of sphere (Vs) = $(4/3)\pi r^3$

Volume of cylinder (Vc) = $\pi r^2 h$: 1 mark

Vs = Vc

r = 9 cm .

Hence, diameter of the cylinder = 18 cm. : 2 marks

OR

Total volume = $(1/3)\pi r^2 h + (2/3)\pi r^3$: 1 mark

= $(726/7)$ cm^3 : 2 marks

Section D

32. Correct table : 1 mark

Correct formula of mode and Median : 2 marks

Mode = 135.76 KWh. Median = 137 KWh. : 2 marks

OR

Correct table : 1 mark

Correct formula of mean : 1 mark

Sum of frequency $f_1 + f_2 = 54$: 1 mark

$f_1 = 34$, $f_2 = 20$: 2 marks

33. For the correct proof of the first part : 2½ marks

For the correct proof of second part : 2½ marks

34. $24/d = \tan 60^\circ$ where d is the distance between the towers. : 1 mark

$d = 24/\tan 60^\circ$

$= 24/\sqrt{3}$ m or $8\sqrt{3}$ m : 1½ marks

$h/d = \tan 30^\circ$ where h is the height : 1 mark

Height of the tower = 8 m : 1½ marks

OR

$\tan 45 = h/d$ where h is the height and d is the distance.

$\tan 60 = (h + r)/d$ where r is the radius : 2 marks

$h = r/(\sqrt{3} - 1)$: 3 marks

35. n^{th} term of an AP is $a_n = a + (n - 1)d$, where a is the first term, d is the common difference. : 1 mark

Ratio of 11^{th} term to 18^{th} term is $(a_{11} = a + 10d)/(a_{18} = a + 17d)$

$a = 4d$: 1½ marks

Sum first n term of an AP is $S_n = (n/2)[2a + (n - 1)d]$: 1 mark

$S_5/S_{21} = \{(5/2)[2a + 4d]\} / \{(21/2)[2a + 20d]\}$

$S_5/S_{21} = 5/49$: 1½ marks

Section E

36. (i). 2 units (1 km) : 1 mark

(ii). 2 units (1 km) : 1 mark

(iii). An isosceles triangle. : 2 marks

37. (i). Angle of elevation = 45° : 1 mark

(ii). They should stand at distance of $14\sqrt{3}$ m : 1 mark

(iii). $20\sqrt{3}$ m : 2 marks

OR

30°

38. (i). 11 : 1 mark

(ii). $25 = 45 + 10d$ or $d = -2$ cm : 1½ marks

Length of the 6^{th} rungs (the middle one) = 35 cm

(iii). Total length of the wood = 9.85 m : 1½ marks

OR

Total length of wood for rungs = 3.85 m